

Rami Al-Belmpeisi

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Education

Technical University of Denmark (DTU)
PhD in Applied Mathematics and Computer Science
Section for Visual Computing.

Copenhagen, Denmark
2023 – 2026

Niels Bohr Institute (NBI), University of Copenhagen (KU)
MSc in Astrophysics
Centre for Star and Planet Formation (StarPlan).

Copenhagen, Denmark
2020 – 2022

Aristotle University of Thessaloniki (AUTH)
BA in Physics
Research Group: Gravitational Waves Group of AUTH.

Thessaloniki, Greece
2015 – 2020

Eberhard Karls Universität (EKUT)
Study Abroad (Erasmus+)
Research Group: Emily Noether Research Team.

Tübingen, Germany
2018 – 2019

Experience

Technical University of Denmark
PhD Researcher

Copenhagen, Denmark
2023 – 2026

Focus I: Advancing Deep Learning & Compute Methodologies

- **Geometric Deep Learning & XAI:** Developed a novel shape-centric Explainable AI framework establishing the first direct mapping between abstract high-dimensional weight spaces and distinct morphological structures. (Methodology submitted to ECCV 2026).
- **Feature Extraction (2D/3D):** Significant experience in both convolutional and transformer-based architectures to extract image features from high-dimensional domains (e.g., 2D images or 3D scalar/vector fields). Features are later used downstream for tasks such as classification, structure detection, and segmentation in both supervised and unsupervised settings.
- **High-Performance Computing (HPPC):** *C++ Optimisation* : Proficient in CUDA for GPU clusters (memory management, kernels) and CPU parallelization (MPI, multi-threading, SIMD, asynchronous communication). *Python Scaling*: Developed efficient CPU workflows (Numba JIT, MPI4Py, NumPy broadcasting) and deployed distributed PyTorch architectures for large-scale simulation processing on GPUs.

Focus II: Black Hole parameter estimation from hot spot kinematics

- **Parameter Retrieval Framework:** Developed a deep learning pipeline to retrieve black hole spin (α) and inclination (i) from simulated hot spot orbits within their Kerr spacetime via 3D GRMHD kinematics simulations.
- **Robustness Testing:** Evaluated model performance against limited orbit visibility (10%–100%) and ability to probe diverse kinematic profiles, including sub-/super-Keplerian velocities and non-equatorial orbits.

Niels Bohr Institute
Researcher & NOT Observing Summer School

Copenhagen, Denmark
2020 – 2022

Focus I: Star Formation & MHD Simulations

- **Hydrodynamic Modeling** : Simulated star formation and early evolution inside turbulent molecular clouds using ideal MHD, performing zoom-in simulations with the oct-based AMR multi-physics code RAMSES (Copenhagen version).

Focus II: Radiative Transfer & Synthetic Observations

- **Synthetic Imaging (RADMC-3D):** Computed observational appearance (synthetic images) by solving the non-local radiative transfer problem for dusty media on RAMSES outputs, accounting for scattering and thermal emission.
- **Bridging Theory & Observation:** Designed a deep learning methodology tailored for protostellar objects, enabling direct comparison between real ALMA photometric observations and synthetic simulation data.

Other Experience

- **Observational Astronomy:** Executed remote observations at the Nordic Optical Telescope (NOT). Managed target selection via visibility curve analysis, optimized signal-to-noise ratios for narrow/broadband filters, and prepared observation blocks. Performed spectroscopic reduction on edge-on galaxies to derive dark matter rotation curves.
- **Planetary Hydrodynamics:** Simulated the photoevaporation of planetary atmospheres under varying UV irradiation regimes. I used the hydrodynamics code "PLUTO" in C++ to model outflows, quantify atmospheric mass-loss rates, and estimate the timescale to reach hydrodynamic stability.
- **High Energy Astrophysics:** Solved relativistic radial stability equations (TOV) to construct neutron star equilibrium models. Analyzed heat transfer mechanisms and convective stability for post-merger remnants using EoS prescriptions derived from full 3D GRMHD simulations.

Publications

- **Al-Belmeisi, R.[†], Yfantis, A. I.[†],** "Deep Learning for Black Hole Parameter Estimation from Hotspot Kinematics." *In preparation for Astronomy & Astrophysics (A&A)*. ([†]Co-first authorship)
- **Al-Belmeisi, R., et al.** "Simulated analogues II: a new methodology for non-parametric matching of models to observations." *Monthly Notices of the Royal Astronomical Society (MNRAS)* 534.4 (2024): 3194-3210.
- **Tuhtan, V., et al.** "Simulated analogues I: apparent and physical evolution of young binary protostellar systems." *Monthly Notices of the Royal Astronomical Society (MNRAS)* 534.4 (2024): 3176-3193.
- **Al-Belmeisi, R., et al.** "Identifying Nonalcoholic Fatty Liver Disease and Advanced Liver Fibrosis from MRI in UK Biobank." *International Workshop on Machine Learning in Medical Imaging (MLMI)*, Springer Nature (2024).
- **Al-Belmeisi, R., et al.** "RESHAPE: Reconstruction-based Explainability for Shapes through importance estimation." *In preparation (Intended for ECCV 2026)*.

Honours & Awards

- **Academic Excellence:** Qualified for university entrance among the top 0.55% of Greek students nationwide.
- **Top Graduate:** Achieved a BSc degree grade of 9.00/10.00, placing me in the top 1% of Physics graduates.
- **Physics/Math:** 12th place in the Greek Annual Physics Contest; Semi-finalist in National Mathematics and Logic Competition.
- **Chess:** 1st place in the Thermaikos Chess Championship for two consecutive years (2014, 2015).

Skills & Interests

Computing & ML: Python (PyTorch, Distributed Training, HPPC), C++, Deep Learning (Explainable AI, Parameter Estimation), 3D Image Analysis, Scientific Computing.

Astrophysics Software: Hydrodynamic Simulation Codes (PLUTO, RAMSES), Radiative Transfer (RADMC-3D).

Interests: Composing Music, Chess, Table Tennis.